at least as measured by light beams, which means that the vacuum fluctuations are exotic.

How this comes about was not deduced in detail until the early 1980s, when Don Page at Pennsylvania State University, Philip Candelas at Oxford, and many other physicists used the laws of quantum fields in curved spacetime to explore in great detail the influence of a hole's horizon on the vacuum fluctuations. They found that the horizon's influence is key. The horizon distorts the vacuum fluctuations away from the shapes they would have on Earth, and by this distortion it makes their average energy density negative, that is, it makes the fluctuations exotic.

Under what other circumstances will vacuum fluctuations be exotic? Can they ever be exotic inside a wormhole, and thereby hold the wormhole open? This was the central thrust of the research effort triggered by Page's noticing that the only way to hold *any* wormhole open is with exotic material.

The answer has not come easily, and is not entirely in hand. Gunnar Klinkhammer (a student of mine) has proved that in flat spacetime, that is, far from all gravitating objects, vacuum fluctuations can *never* be exotic—they can never have a negative average energy density as measured by light beams. On the other hand, Robert Wald (a former student of Wheeler's) and Ulvi Yurtsever (a former student of mine) have proved that in curved spacetime, under a very wide variety of circumstances, the curvature distorts the vacuum fluctuations and thereby makes them exotic.

Is a wormhole that is trying to pinch off such a circumstance? Can the curvature of the wormhole, by distorting the vacuum fluctuations, make them exotic and enable them to hold the wormhole open? We still do not know, as this book goes to press.

In early 1988, as theoretical studies of exotic material were getting under way, I began to recognize the power of the kind of research that Carl Sagan's phone call had triggered. Just as among all *real* physics experiments that an *experimenter* might do the ones most likely to yield deep new insights into the laws of physics are those that push on the laws the hardest, then similarly, among all *thought* experiments that a *theorist* might study, when probing laws that are beyond the reaches of modern technology, the ones most likely to yield deep new insights are those that push the hardest. And no type of thought experiment pushes the laws of